Chapter 21: **Computers in Graph Theory**

We briefly review ways in which computer support might be used in graph theory research.

21.1 Data from Small Graphs

Researchers have compiled databases of all small graphs up through a certain order. For example, all graphs, all trees, all planar graphs, and all regular graphs. So one can try out a conjecture on small graphs or just get data that might suggest a pattern.

Specific Example: The independent domination number of cubic graphs. We [2] ran a database search asking for the maximum independent domination number of a cubic graph of specific order. The biggest value relative to the order we found was $\frac{3}{8}$ (apart from tiny cases such as $K_{3,3}$). In fact there were two cubic graphs on 16 vertices that have $i = 6$, and two cubic graphs on 8 vertices that have $i = 3$. Using software to draw these graphs suggested a pattern where the order-16 graphs are made up of two order-8 pieces. So, this suggested a construction of graphs on $8m$ vertices. Now, to prove that the general graph does not have a smaller independent domination set, again a computer was harnessed to verify that dominating or almost dominating sets on each piece have suitable properties (though in theory this could be done by hand).

21.2 Exhaustive Search

Exhaustive search uses computers to try all possibilities in order to show that something does not exist.

A classic example is trying to find Ramsey numbers. That is, show that one can or cannot color the edges of K_n without having a red K_r or blue K_s.

Specific Example: The bipartite Ramsey number $b(2, 2, 2)$. That is, how large can $K_{m,m}$ be and one still color the edges with three colors such that no monochromatic $K_{2,2}$. In particular, we [1] wanted to know if it was possible to 3-color the edges of $K_{10,10}$ to avoid monochromatic 4-cycle. Theory showed that there is a unique bipartite graph H on 10 vertices and 34 edges without a 4-cycle. So one can hope to fit two copies of H in, and then check that what remains is also 4-cycle-free. And computer did this. Except, that the problem was already solved by Exoo twenty-five years before...

Much of the hard work in exhaustive search goes into exploiting symmetry to reduce the number of cases. One of the major tools is McKay's `nauty` program. Indeed, it was used in finding some of the databases mentioned above.

Specific Example: In original Ramsey numbers, circulants were the extremal cases for small numbers, so perhaps they are in general. Thus, one might need to know the independence number of a circulant. Symmetry kicks in: since graph is vertex-transitive, one can start with any vertex v. Can assume the minimum displacement is from v to the next vertex clockwise, by rotation symmetry. If we are looking for an independent set of specific size, can assume that half are on the left, by mirror symmetry, and so on.

A more recent development is the idea of converting the problem to logic and then using a SAT solver.

21.3 Other Ideas

1. ***Heuristic search.*** This is used to try to find some graph or configuration, using search techniques developed in artificial intelligence. One example was for certain domination problems on grids, such as attacking all squares with queens.

2. ***Specific Programming.*** One of the first proofs to use computer programming was the proof of the 4-color theorem. Essentially that proof uses induction. Specifically, a reducible configuration is a subgraph that one can shrink in some way, apply induction, and recolor the whole thing. So the computer is used (a) to find these configurations, and then (b) to show that every planar graph must have one of them (unavoidable).

3. ***Optimization.*** Tools such as linear programming, integer programming, and computer algebra for calculus, are also useful.

21.4 References

[1] Bipartite Ramsey numbers and Zarenkiewicz bounds, W. Goddard, M.A. Henning, and O. Oellermann. Discrete Math., 219 (2000), 85–95.

[2] On the independent domination number of regular graphs, W. Goddard, M.A. Henning, J. Lyle, J. Southey. Ann. Comb. 16 (2012), no. 4, 719–732.